

Directional Derivative and the Gradient Vector

Trim Ch 11, S. 12.8, 12.4

Outline:

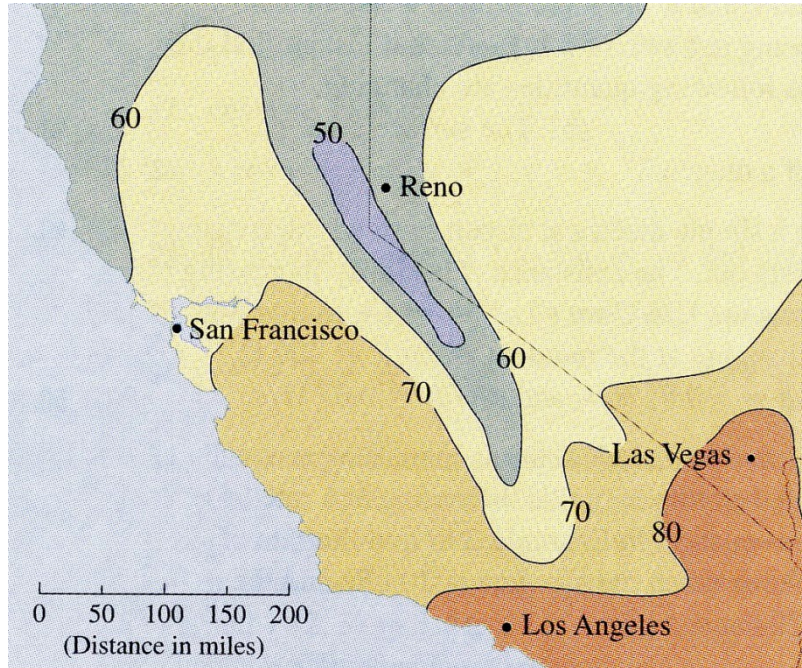
1. Directional derivative and the gradient vector
2. Properties of the directional derivative and the gradient vector



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Weather Maps

Level curves (or isotherms)
join locations with the
same temperature

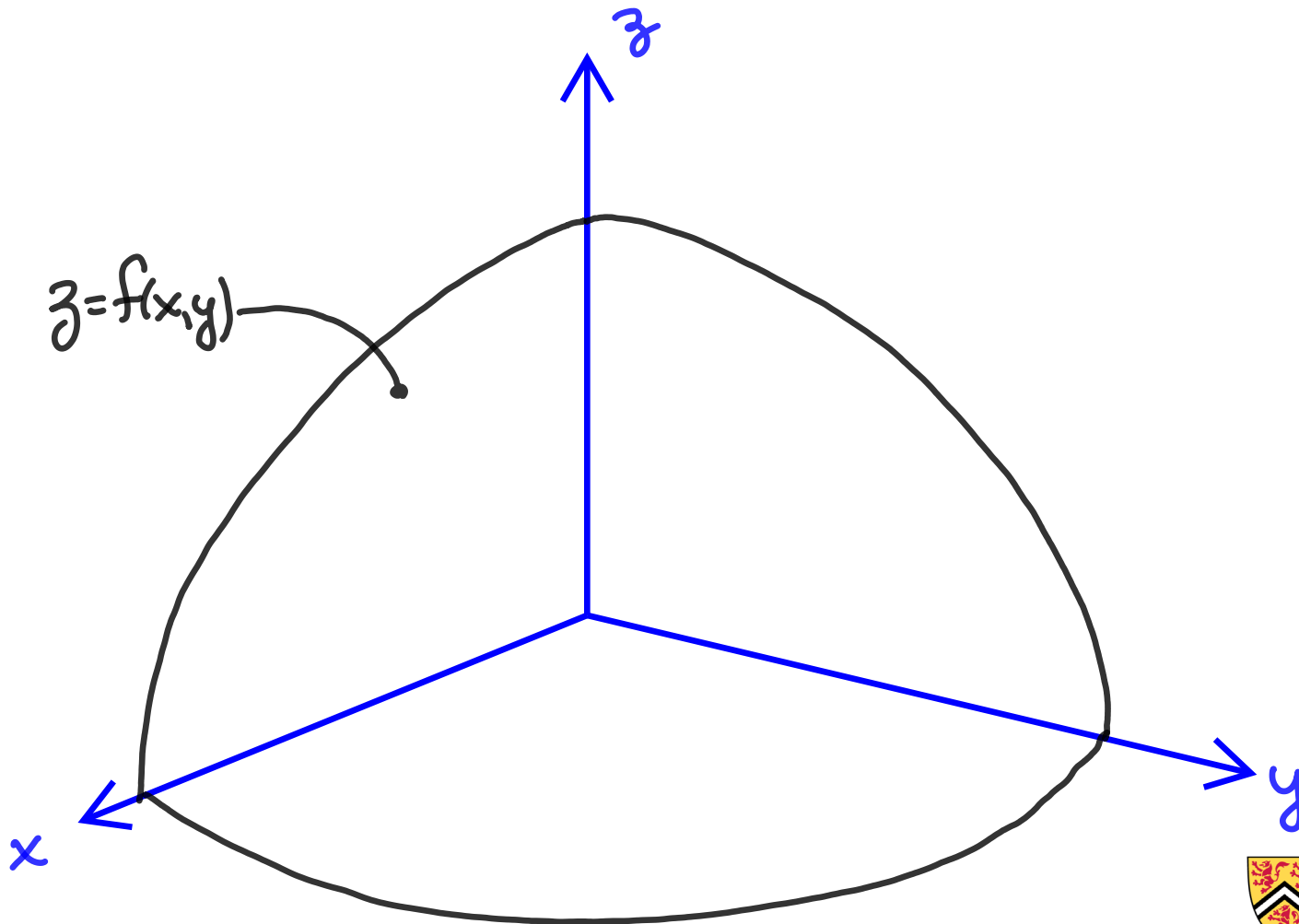


Contour map of the temperature function $T(x,y)$ for the states of California and Nevada at 3:00pm on October 10, 1997.

At a location such as Reno, what if we want to know the rate of change of temperature when traveling southeast?

Directional Derivative

Given the surface $f(x, y)$, suppose we want to step off in the direction of an arbitrary vector $\hat{u} = (u_1, u_2)$ of increasing “ s ”



Directional Derivative

If the parameter “s” measures the length of the line in the direction of $\hat{u}$, the equation of this line can be written as,

As the point Q gets closer to the point P_0 , we can calculate the slope of the tangent line (or rate of change) of $f(x, y)$ at P_0 stepping off in the direction of $\hat{u}$ as,

In the limit, this is called the Directional derivative of $f(x, y)$ at P_0 in the direction of $\hat{u}$.



Directional Derivative and the Gradient Vector

Let us now develop an efficient formula to calculate the directional derivative using the chain rule,

Directional Derivative and the Gradient Vector

The directional derivative of a function $f(x, y)$ at P_0 , stepping off in the direction of $\hat{u}$ can therefore be calculated from,

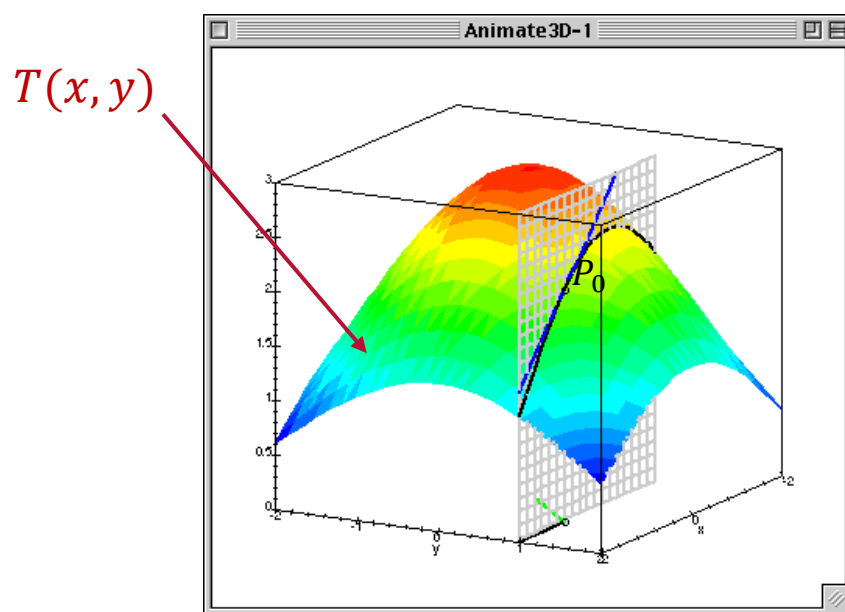
where,

$$\vec{\nabla} f = \left(\frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} \right)$$

is called the gradient vector.

Directional Derivative – Physical Interpretation

If $T(x, y)$ is the temperature at each point $P_0(x_0, y_0)$ of the surface, defined over a region R ,



The directional derivative is the **instantaneous rate of change** of temperature at P_0 **stepping off in the direction of $\hat{u}$**

$$D_{\hat{u}}T|_{P_0} = \left. \frac{dT}{ds} \right|_{\hat{u}, P_0} = \vec{\nabla}T|_{P_0} \cdot \hat{u}$$



Example:

Find the directional derivative $D_{\hat{u}}f(x, y)$ at $(1, 2)$ of the function,

$$f(x, y) = x^3 - 3xy + 4y^2$$

where $\hat{u}$ is the unit vector given by the angle $\theta = \frac{\pi}{6}$

Solution:

Example:

Properties of the Directional Derivative and the Gradient Vector

Directional Derivative in the Direction of $\hat{u}$

The directional derivative of a function $f(x, y)$ at P_0 in the direction of $\hat{u}$ gives us insight into some of the properties of the gradient vector,

$$D_{\hat{u}}f|_{P_0} = \left. \frac{df}{ds} \right|_{\hat{u}, P_0} = \vec{\nabla}f|_{P_0} \cdot \hat{u}$$

The directional derivative $D_{\hat{u}}f|_{P_0}$ is the component of the gradient vector at P_0 in the direction of $\hat{u}$.

Summary – Part I

- **The directional derivative is the instantaneous rate of change of a function $f(x, y)$ at P stepping off in the direction of $\hat{u}$**
- The directional derivative is the component of the gradient vector in the direction of $\hat{u}$